Miaomiao Chen

Research Scientist @ AstraZeneca PLC

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SUMMARY

Ph.D. in Chemistry with 7+ years of experience in machine learning modeling and large-scale data analysis, grounded in statistics and applied mathematics. Hands-on experience building LLM-powered and Agentic AI systems, with practical expertise in EDA, data pipelines, and production MLOps. **Fun fact**: National Chess Champion (multiple titles, China).

EDUCATION

Georgia Institute of Technology, MS in Computer Science (ML Track) — 2024 – Present
University of Virginia, PhD in Chemistry — 2024
ShanghaiTech University, BS in Bioscience — 2019

SKILLS

- **Programming:** Python, SQL, Spark
- **Machine Learning:** Transformers, LLMs, RL, Agentic AI (LangChain), PyTorch, RecSys (RAG, HSTU)
- **MLOps & Deployment:** EDA, Domino (Kubernetes), FastAPI services, Bazel

EXPERIENCE

Research Scientist, Clinical Proteomics, AstraZeneca Pharmaceuticals PLC, MD — Jul 2024 – Present

- Developed a LangChain-based RAG service integrated with MonetDB, implementing query orchestration and routing to downstream analytical pipelines, uncovering proteomics patterns across AstraZeneca's clinical drug portfolio.
- Built multi-variant ML models, engineering custom patient embeddings using NMF and Two-Tower architectures within a production MLOps pipeline for drug-response prediction, achieving 0.82 ROC-AUC on clinical datasets.
- Fine-tuned an encoder–decoder foundation model via transfer learning on proteomics data to predict drug response rates within clinical trials, achieving a +0.20 ROC-AUC improvement.
- Deployed production Python Flask applications on Domino (Kubernetes) to automate clinical data quality control with statistical analysis and visualization pipelines, now a core part of the group's workflow.

PhD Researcher, University of Virginia, VA — 2019 – 2024

- Designed ML models (RF, XGBoost, LSTM) for drug candidate down-selection, achieving 85.3% accuracy, shortening experimental timelines by months, and identifying 2 of 10 effective candidates, with results published in *JACS*.
- Performed large-scale EDA and designed A/B testing for anti-inflammation drug studies, contributing to publications including first-authored work in *PNAS* (Proceedings of the National Academy of Sciences).

SELECTED PROJECTS

Krono: Voice-Activated AI News Agent — 2025 – Present

- Designed a LangGraph-based multi-agent system for intent detection, tool routing, retrieval, and generation, supporting real-time, fully voice-driven news exploration with persistent conversational context and natural follow-ups.
- Implemented a lightweight, thread-based audio interaction architecture enabling real-time interruption and refinement (<100 ms response), achieving high concurrency and ∼60% lower memory usage than multiprocessing approaches.
- Built and launched a production voice-based AI news agent that performs real-time retrieval, ranking, and summarization of market news, reducing user lookup time by ∼70%.
- Post-trained Qwen-0.5B using GRPO for large-scale news classification, cutting external LLM API usage by 60% and reducing inference cost by 50%, while integrating SenseVoice ASR, GLM-4-Flash, and financial data APIs.

Generative Retrieval-based Video Recommendation System — 2024 – Present

- Designed and implemented a SOTA transformer-based HSTU recommender for UGC short-video recommendation, achieving +11% HR@k / NDCG@k improvement over SASRec via stronger sequential modeling.
- Integrated point-wise generative next-item embedding prediction and customized RoPE with timestamp- and duration-aware encoding, delivering a 15% NDCG uplift and reducing reliance on negative sampling.
- Introduced self-adaptive rating normalization based on watch-ratio signals and built an end-to-end training/evaluation pipeline on KuaiRec, enabling robust benchmarking and an additional ∼3% performance gain.

Reinforcement Learning of Flappy Bird Game — 2024 – 2025

- Led the design of a DQN-based reinforcement learning pipeline for training an autonomous Flappy Bird agent, incorporating experience replay and ε-greedy exploration to stabilize learning.
- Implemented Double and Dueling DQN architectures, accelerating reward convergence by 80% and significantly improving training stability over vanilla DQN.
- Built a configurable, hyperparameter-driven RL framework (YAML-based) adaptable across environments, enabling the agent to pass 300+ obstacles and sustain ∼16.5% average reward improvement between learning milestones.